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Novel Mechanistic Insights in Radiation Biology (2019–2024)

Radiation biology has witnessed a surge of discoveries in the past five years, revealing new molecular mechanisms that govern cellular responses to ionizing radiation. DNA double-strand breaks (DSBs) remain the most lethal lesions induced by radiation, and their repair involves intricate networks of proteins and signaling pathways [1]. Beyond the classical DNA repair proteins, recent studies highlight novel protein–protein interactions, regulatory complexes, and cellular processes that modulate DNA repair efficacy, oxidative stress responses, and cell fate after radiation exposure. Here, we review **20 of the most interesting and novel mechanistic ideas in radiation biology research** (2019–2024), with an emphasis on human biology (while drawing insights from other organisms where relevant). These findings span newly discovered DNA repair complexes, cross-talk between DNA damage and other cellular pathways (e.g. RNA metabolism, immune sensing, redox regulation), and factors underlying radiation tolerance or sensitivity. Each section below elaborates a distinct emerging concept, citing primary literature and preprints (including bioRxiv) to provide a rigorous, up-to-date overview for experts in the field.

1. The Shieldin Complex: A New NHEJ Effector in DNA Repair

One of the landmark discoveries was the identification of *Shieldin*, a four-subunit protein complex that acts as a 53BP1-dependent effector of non-homologous end joining (NHEJ). Shieldin was first described in 2018 as a complex comprising REV7 (also known as MAD2L2) and three previously uncharacterized proteins (named RINN1, RINN2, RINN3) [2]. Recruitment of Shieldin to DSBs via the ATM–RNF8–RNF168–53BP1–RIF1 axis allows it to bind ssDNA at resecting DNA ends, thereby *"shielding"* the DNA ends from excessive nucleolytic resection [2] [3]. This restraint of end-resection favors classical NHEJ over homologous recombination. Functionally, Shieldin loss impairs intrachromosomal NHEJ and causes resistance to PARP inhibitors in BRCA1-deficient cells, whereas Shieldin is required for 53BP1-mediated DNA end protection during class-switch recombination in B cells [2] [4]. Notably, Shieldin also impacts cellular radiosensitivity: by limiting DNA end resection and promoting efficient ligation, Shieldin contributes to the repair of radiation-induced DSBs and enhances cell survival after ionizing radiation [5] [6]. In summary, the discovery of Shieldin has filled a gap in our understanding of how 53BP1's pathway enforces NHEJ, identifying a new protein–protein complex at DNA breaks that determines repair pathway choice and influences radiation response.

2. Polymerase θ-Mediated End Joining and RNA-Templated Repair

DNA polymerase theta (Polθ, encoded by *POLQ*) has emerged as a critical player in alternative end-joining (alt-EJ or TMEJ), especially in the context of DNA repair and radiation resistance. Polθ is a unique enzyme with both polymerase and helicase domains, notable for its error-prone DNA synthesis and ability to pair microhomologies on resected DNA ends [7] [8]. In the last five years, multiple studies have elucidated new functions and interactions of Polθ:

- **Polθ in mitotic DSB repair:** Gelot *et al.* (2023) demonstrated that Polθ can operate in **mitosis**, a phase when classical NHEJ and homologous recombination are inactive [9]. Polθ is **activated by Polo-like kinase 1 (PLK1)** phosphorylation in mitosis and recruited to broken DNA ends through an interaction with the BRCT domains of TOPBP1 [9]. There, Polθ mediates end-joining of mitotic DSBs, preserving genome integrity in cells that enter mitosis with unrepaired breaks. Loss of Polθ leads to defective mitotic DSB repair and lethal genomic instability, especially in HR-deficient cells [10]. This revealed a new cell-cycle-specific role of Polθ and explains the synthetic lethality observed between Polθ inhibition and homologous recombination defects [11].

- **Polθ as a reverse transcriptase:** Remarkably, a 2021 study showed that Polθ can utilize RNA as a template for DNA repair synthesis [12]. **Pomerantz and colleagues (Science Advances, 2021)** reported that human Polθ exhibits efficient **reverse transcriptase activity**, inserting dNTPs on an RNA template with higher speed and fidelity than on DNA [13]. The crystal structure of Polθ engaged with an RNA/DNA hybrid revealed a major conformational change accommodating A-form RNA/DNA [14]. Importantly, Polθ was shown to promote **RNA-templated DNA repair in cells**, suggesting that RNA transcripts (such as those generated at DSB sites) could serve as templates to fill in missing DNA information [15]. This finding uncovers a novel mechanism of genomic maintenance: Polθ may mitigate damage by copying sequence from RNA, analogous to retrotransposon or viral reverse transcriptases.

- **Targeting Polθ for radiosensitization:** Because Polθ is often upregulated in tumors and helps them survive genotoxic stress [16] [17], it has become a target for therapy. Inhibitors of Polθ have been developed, and recent work indicates that Polθ disruption preferentially radiosensitizes cancer cells with HR defects [18]. For example, using Polθ inhibitors in combination with radiation or ATR inhibitors can drive tumor cells into catastrophic repair failure while sparing normal cells that rely more on HR [19] [20]. This therapeutic angle, while beyond our mechanistic focus, underscores Polθ's centrality in tolerating radiation damage.

In summary, Polθ's role has expanded from a mere alt-EJ polymerase to a multifaceted enzyme active in unusual contexts—mitotic repair and even RNA-templated DNA synthesis—highlighting a unique protein–protein interplay (Polθ–PLK1–TOPBP1) and a DNA/RNA substrate flexibility that together enhance cellular survival after radiation damage.

3. Liquid-Liquid Phase Separation in DNA Repair Foci

A striking paradigm shift in how we view DNA damage foci is the discovery that certain DNA repair proteins form **phase-separated condensates** at sites of damage. *Liquid-liquid phase separation (LLPS)* refers to the ability of biomolecules to demix and form dynamic, liquid droplet-like compartments. Two seminal findings illustrate this concept:

- **53BP1 condensates:** In 2019, Kilic *et al.* showed that 53BP1, a key DNA damage response (DDR) scaffold, forms liquid-like condensates at DSB sites [21]. Using live-cell microscopy and biophysical assays, they demonstrated that 53BP1 foci exhibit hallmarks of liquid droplets: spherical shape, ability to fuse, sensitivity to disruption by 1,6-hexanediol or heat, and even in vitro phase separation of purified 53BP1 [21] [22]. Notably, upstream repair factors (γH2AX, MDC1) are required to initially recruit 53BP1, but only 53BP1 (not MDC1) showed this droplet behavior [23]. This suggests a two-stage assembly: first, a precise protein accumulation at DNA lesions, and second, condensation of 53BP1 into a liquid compartment that can concentrate repair and signaling proteins.

- **Functional impact of 53BP1 LLPS:** The phase-separated 53BP1 domains were found to serve more than just local repair – they coordinate *global signaling*. Specifically, 53BP1 condensates help sequester and activate the tumor suppressor p53. Cuella-Martin *et al.* observed that p53 can accumulate in a subset of 53BP1 droplets, and that disrupting 53BP1's ability to phase-separate (by mutating its intrinsically disordered regions) impairs p53 stabilization and p21-dependent cell-cycle arrest after DNA damage [24] [25]. Thus, 53BP1's liquid phase not only retains broken ends for repair but also amplifies checkpoint signaling (linking local DNA damage to global cell fate decisions). The idea that a *"liquid scaffold"* at a DNA break coordinates both repair and transcriptional response is a novel perspective in radiation biology.

- **FUS and early DDR condensates:** Beyond 53BP1, other RNA-binding proteins such as FUS (an RBP with prion-like domains) also contribute to DDR condensates. **Levone *et al.* (J Cell Biol, 2021)** showed that **FUS undergoes LLPS at DNA damage sites and is required for the *initiation* of DDR signaling** [26]. In FUS-knockout cells, recruitment of key repair and signaling factors to laser-induced DNA breaks was impaired [26]. FUS's phase-separating ability appears to help organize the very early DNA damage response, perhaps by forming a nucleation hub for other factors. These findings emphasize that *LLPS is a critical biophysical mechanism underpinning DNA repair foci formation and function.* Disordered protein domains and transient multivalent interactions (common in DDR proteins) drive these phase transitions.

In summary, the concept of DNA damage sites as membraneless organelles formed by phase separation is a major innovation. It adds a new layer of spatial-temporal regulation in the nucleus: radiation-induced breaks can seed droplet-like domains that concentrate repair proteins and signaling molecules, thereby accelerating repair and orchestrating downstream outcomes [27] [21]. Understanding the molecular grammar of these condensates (which sequences or modifications promote LLPS) is now a vibrant area, with implications for why certain mutations (e.g., in 53BP1 or FUS) might disrupt genome stability and cell viability after radiation.

4. R-Loops and RNA-Binding Proteins in the DNA Double-Strand Break Response

It is becoming increasingly clear that **RNA molecules and RNA-binding proteins (RBPs)** are integral to the DSB response, far beyond their traditional roles in gene expression. An important structure in this context is the *R-loop*, a three-stranded nucleic acid structure consisting of an RNA–DNA hybrid and a displaced single DNA strand. R-loops can form during transcription or at DNA breaks and have emerged as double-edged swords in genome stability. Key insights from recent studies include:

- **R-loops at DNA breaks:** Evidence suggests that RNA polymerase II can produce transcripts near DSBs (discussed further in Section 5), which can hybridize with DNA to form R-loops. While some R-loops at breaks may serve regulatory purposes, uncontrolled R-loops are generally dangerous, as they can stall replication and impede proper repair. Cells have dedicated factors to resolve R-loops;

for example, helicases like DDX5 and senataxin (SETX) and nucleases like XRN2 degrade the RNA in R-loops. A recent study showed that **DDX5 resolves R-loops at DSB sites to promote accurate repair and prevent deletions** [28]. Loss of DDX5 led to persistent R-loops and abnormal repair outcomes, underscoring that timely R-loop removal is crucial for DSB repair fidelity.

- **RBPs as DNA repair modulators:** A "flurry of studies" has identified many classical RBPs moonlighting in the DSB response [29]. These include heterogeneous nuclear RNPs (hnRNPs), serine/arginine-rich (SR) proteins, and proteins of the DNA/RNA-binding *DBHS* family (such as NONO and SFPQ). Such RBPs are often phosphorylated by ATM/ATR kinases upon DNA damage [30], suggesting they are recruited into the DDR. **Mechanistically, RBPs influence DSB repair in several ways** [31]: (i) they scaffold or recruit DNA repair factors (for instance, NONO can partner with Ku to aid NHEJ), (ii) they protect or process RNA transcripts at breaks (e.g., by preventing harmful RNA:DNA hybrids), (iii) they participate in DNA damage signaling cascades (some RBPs bind and modulate kinases or ubiquitin ligases in the DDR), and (iv) they help organize phase-separated repair compartments (many RBPs have disordered regions that contribute to condensate formation, as in the case of FUS). An illustrative example is **CIRBP (Cold-Inducible RNA Binding Protein)** – typically a stress-response RBP – which upon irradiation relocalizes to DSBs and assists in checkpoint signaling [32] [33]. Another example is **TIA-1** and **TIAR**, translational regulators that accumulate at DSBs and modulate repair pathway choice, partly by binding repair transcripts.

- **Non-coding RNAs as guides and tethers:** RBPs also interact with various non-coding RNAs induced by damage. These RNAs (discussed below) can remain tethered to break sites, and RBPs like **RAD52** and **BRCA1** have been proposed to bind RNA and mediate RNA–DNA hybrid annealing in some repair contexts [34]. The RBPs **Ku70/Ku80** (traditionally DNA-binding) even have an RBP facet: they can bind an RNA component of telomerase at chromosome ends and possibly RNA at breaks, hinting at RNA's broad influence.

In summary, **R-loops and RBPs introduce an RNA dimension to radiation responses**. Uncontrolled R-loops pose a threat by stalling replication and inducing replication stress, which itself leads to more DSBs. Consequently, cells deploy R-loop-resolving factors (like DDX helicases and RNases) to maintain genome stability [35] [28]. Simultaneously, many RBPs are repurposed in DDR to either remove these RNA structures or to exploit them for signaling and repair complex assembly. This emerging picture blurs the line between the transcriptome and the genome maintenance machinery, illustrating that **cellular responses to radiation are not purely DNA-centric but involve RNA-based mechanisms and RBP networks** [29] [36].

5. Damage-Induced Transcription and RNA-Templated DNA Synthesis

Alongside the roles of existing RNAs and RBPs, another revolutionary idea is that cells *actively transcribe new RNA at sites of DNA damage*, and that these transcripts can function in the repair process. This challenges the longstanding view that transcription near a DSB is uniformly silenced; instead, it appears that in some contexts, **RNA polymerase II is deliberately recruited to DSBs to produce damage-responsive RNAs**.

- **DSB-induced lncRNA (dilncRNA) synthesis:** Early hints came from studies in yeast and mammals showing RNA Pol II at break sites and the presence of non-coding RNAs stemming from the broken locus. In 2021, Sharma *et al.* provided direct evidence that the **MRN complex (MRE11–RAD50–NBS1)** is capable of recruiting RNA Pol II to DNA ends and initiating transcription of "damage-induced long non-coding RNA" (dilncRNA) [37]. They reconstituted a minimal system *in vitro* and found that recombinant MRN plus RNAPII could drive transcription from a DSB end [37]. Interestingly, MRN's DNA endonuclease activity was not required; rather, its ability to melt/open the DNA end was key to allowing RNAPII to start RNA synthesis [38]. Single-molecule assays showed MRN can unwind the DNA terminus, and mutants that couldn't unwind failed to support transcription [38]. The addition of RPA (which binds ssDNA) further enhanced this MRN-driven transcription [39]. These findings support a model in which **MRN generates a single-stranded DNA tail at the break and then stabilizes RNAPII there to transcribe a nascent RNA** [38]. The transcripts can originate from both break ends and may remain localized.

- **Functions of damage transcripts:** What do these dilncRNAs do? Growing evidence suggests they contribute to DDR signaling and repair accuracy. They can base-pair with resected DNA or with other RNAs to form RNA:DNA hybrids at the break. Such hybrids might aid in bridging two ends of a DSB (by complementary annealing of transcripts from each end) or recruit factors that bind RNA:DNA hybrids (like RAD52 or ATR). Additionally, the process of transcription itself may loosen chromatin or push off bound proteins, helping resection or remodeling. Some transcripts are processed into shorter RNAs (sometimes called DDRNAs or DNA-damage response RNAs) that have been proposed to recruit ATM/ATR or other factors to the break [40] [41]. However, an overabundance of RNA at breaks can be harmful (tying back to Section 4 on R-loops). Thus, the cell likely tightens the spatial and temporal window of damage-transcription and then removes the RNA.

- **RNA-templated DNA repair by Polθ:** Perhaps the most astonishing related discovery, as mentioned in Section 2, is that Polθ can use these damage-induced RNAs as templates for repair DNA synthesis [12]. In cases where a DSB has resected over a gap, an RNA transcribed from one side of the break could anneal to the complementary side and Polθ would copy it to fill the gap – effectively *"information recovery"* via an RNA intermediate. **Mallon *et al.* (2021) demonstrated that Polθ's reverse transcriptase activity enables such RNA-templated repair in human cells** [15]**.** This mechanism might be especially relevant when the homologous template is absent and NHEJ is inefficient – a scenario common in heterochromatin or repetitive regions where a quick fix is needed and transcripts are available as templates.

In summary, these findings reshape our understanding of the central dogma under stress conditions: **RNA Pol II and transcription are not mere victims to DNA damage but active participants in the repair process**. MRN's role in melting DNA ends and recruiting Pol II [37], combined with Polθ's ability to turn RNA back into DNA [12], uncovers a circular flow of information (DNA→RNA→DNA) at the break site. It's an elegant solution to restore lost sequence and signal the presence of damage. For radiation biologists, this means that modulating transcription or RNA processing at breaks could influence repair efficiency – an entirely new angle on radioprotection or sensitization.

6. cGAS–STING DNA-Sensing Pathway Activation by Radiation

Ionizing radiation not only damages nuclear DNA but can also produce cytosolic DNA fragments (from chromosomal pulverization or leakage) that trigger innate immune pathways. The **cGAS–STING pathway** is a cytosolic DNA sensing mechanism normally tasked with detecting viral DNA. In the context of radiation, it has gained prominence for linking DNA damage to immune activation and inflammation. Key insights include:

- **Micronuclei as radiation byproducts:** Cells exposed to radiation or other genotoxic stress often form *micronuclei* – small, extranuclear bodies containing chromosome fragments or whole chromosomes that were not properly segregated. Micronuclei have a tendency to undergo nuclear envelope rupture, exposing their DNA to the cytoplasm. cGAS (cyclic GMP–AMP synthase) can bind this DNA and produce the second messenger cGAMP, which activates STING (Stimulator of Interferon Genes) on the endoplasmic reticulum, leading to IRF3 phosphorylation and type-I interferon (IFN) gene expression [42] [43]. This chain can result in an anti-tumor immune response (if in cancer cells) or, conversely, chronic inflammatory damage (in normal tissue).

- **Reevaluation of cGAS activation by micronuclei:** While earlier reports suggested that radiation-induced micronuclei robustly activate cGAS–STING, a **2024 study by Takaki *et al.*** in *Molecular Cell* yielded a surprising result. They found that **micronuclei formed by ionizing radiation or replication stress *do bind cGAS, but often fail to fully activate the STING pathway*** [44]. The authors observed cGAS localization to ruptured micronuclei, yet no downstream signaling – no cGAMP production, no STING/TBK1/IRF3 phosphorylation, and no interferon-stimulated gene (ISG) induction [44]. The reason was traced to **nucleosomes in the micronuclear DNA**: micronuclei contain densely packed chromatin (histone-bound DNA), which inherently *inhibits cGAS enzymatic activation* [45] [46]. In essence, cGAS binds the DNA but is unable to oligomerize efficiently on nucleosomal substrates, thus staying in an inactive state. Only when micronuclei catastrophically break down into naked DNA does full cGAS activation occur. This nuanced finding suggests that cells might tolerate a certain level of extranuclear DNA without mounting an IFN response, and it has implications for radiation dosing and timing in immunotherapy (since not all DNA damage translates to immunogenic signal).

- **Mitochondrial DNA and STING:** Another source of cytosolic DNA after irradiation is **mitochondrial DNA (mtDNA)**. Radiation can damage mitochondria, leading to mtDNA release into the cytosol. Recent work indicates mtDNA may be a major trigger for cGAS–STING in some contexts [47]. Yamazaki *et al.* (2020) showed that when autophagy is inhibited (see Section 7), irradiated cells accumulate cytosolic mtDNA and this leads to robust STING-dependent interferon secretion [48] [49]. In tumors, this phenomenon can generate *abscopal effects* – local radiation causing systemic anti-tumor immunity facilitated by STING-driven T cell activation. Indeed, a genetic variant in the human **STING1 gene** has been associated with differential normal tissue radiosensitivity (radiogenomic evidence reviewed in Section 13), underscoring the pathway's clinical relevance [50].

- **Therapeutic angle:** The cGAS–STING pathway is being harnessed to improve radiotherapy outcomes. Preclinical studies combine radiation with STING agonists to boost tumor immunogenicity [51]. Conversely, in normal tissue protection, transient STING pathway inhibitors might mitigate radiation-induced inflammatory damage (e.g. pneumonitis or proctitis). The paradox is that while STING activation can help clear tumor cells via immune mechanisms, excessive or inappropriate activation in healthy cells can lead to fibrosis or autoimmunity. The 2024 finding that nucleosomal DNA in micronuclei dampens cGAS activation [45] might explain why not every irradiated cell launches inflammation – it offers a molecular checkpoint that distinguishes "normal" DNA (chromatinized, in micronuclei) from "foreign" DNA (naked, as in viruses or ruptured nuclei).

In summary, **radiation-provoked activation of cGAS–STING is a prime example of how DNA damage intersects with innate immunity**. Cutting-edge studies refine our understanding: while the presence of cytosolic DNA is necessary, its form (nucleosomal or not) and the cell's ability to clear it (via autophagy or other means) determine the outcome [46] [48]. This area beautifully bridges radiation biology with immunology, explaining phenomena like the abscopal effect and suggesting why patients with certain immune genotypes might experience differing side effects or tumor control from radiotherapy.

7. Autophagy and Mitophagy Modulation of Radiation Responses

Autophagy, the process by which cells degrade and recycle their own components, has emerged as a significant modifier of radiation damage outcomes. In particular, **mitophagy** (selective autophagy of mitochondria) plays a complex role in how cells cope with radiation-induced injury. Recent studies reveal that autophagy can have dual influences – sometimes protective to normal cells, other times enabling cancer cell survival or affecting immune signaling:

- **Autophagy suppresses cytosolic DNA signaling:** Autophagy is a major route for disposing of damaged organelles and macromolecules. When DNA damage occurs, autophagy can remove fragments of DNA or whole chromosomes that escape the nucleus, as well as damaged mitochondria that might leak DNA. **Yamazaki *et al.* (2020)** demonstrated that autophagy inhibition in irradiated cells led to increased cytosolic DNA (particularly mtDNA) and a consequent spike in type-I interferon via cGAS–STING [48]. Autophagy-proficient cells, in contrast, sequestered or degraded this DNA, blunting the immune response. In mouse tumor models, autophagy-deficient tumors showed greater radiation-induced inflammation and immune-mediated tumor clearance (an enhanced abscopal effect) [52] [49]. Thus, autophagy in tumor cells can be a *radioresistance mechanism*, dampening immunogenic signals (like IFN) and thereby allowing tumors to evade the immune system after radiation. This has led to the concept of combining autophagy inhibitors with radiotherapy to *boost* anti-tumor immunity by unleashing more cytosolic DNA and danger signals.

- **Mitophagy and DNA damage levels:** Counterintuitively, autophagy – by removing damaged mitochondria – might protect normal cells but also *reduce* the efficacy of radiotherapy in killing tumor cells. A 2023 study by Ren *et al.* found that **radiation actively triggers mitophagy, and that enhancing mitophagy can increase DNA damage in the nucleus** [53] [54]. The proposed mechanism is that mitophagy initiation acts as an upstream signal that modulates cell cycle checkpoints and DNA repair. When mitophagy was upregulated (via overexpression of Parkin or BNIP3, key mitophagy proteins), irradiated cells showed *less G2/M arrest and more unresolved DNA breaks* (higher γ-H2AX, 53BP1 foci) [53]. The cells proceeded through division with damage, leading to increased cell death. In a melanoma model, boosting mitophagy before irradiation actually improved tumor control and animal survival [55]. This suggests a potential therapeutic strategy: transiently *increase* mitophagy in cancer cells to push them past checkpoints with lethal DNA damage. It also highlights a critical point – **mitochondrial status can feed back to nuclear DNA repair**. Damaged mitochondria produce excess ROS and can stall cell cycle; removing them via mitophagy might paradoxically deprive the cell of a pause needed for repair, leading to more DNA damage accumulation.

- **Radiation and general autophagy flux:** Clinically, radiation can induce autophagy in many cell types. This is often seen as a survival response – cells use autophagy to clear damaged proteins and organelles and to generate energy for recovery. High-dose radiation, however, might overactivate autophagy to the point of *autophagic cell death*. The balance is delicate. **Chemical or genetic autophagy inhibitors (like chloroquine or ATG gene knockdowns)** generally radiosensitize tumor cells (consistent with them no longer being able to clear damage), but in normal tissues autophagy is needed to mitigate injury (e.g., protect intestinal stem cells from radiation syndrome by clearing damaged mitochondria).

In summary, **autophagy/mitophagy is a key moderator of radiation responses, linking subcellular organelle integrity with DNA repair and immune signaling**. Enhanced autophagy in a cell can mean improved cleanup of radiation damage but at the potential cost of hiding danger signals from the immune system [56] [57]. On the other hand, tweaking mitophagy can drive cells into lethal misrepair by altering checkpoint dynamics [53]. These findings encourage a context-dependent approach: transient autophagy inhibition in tumors (to heighten DNA damage and immunogenicity) versus autophagy promotion in normal tissues (to alleviate radiation toxicity). The interplay between autophagy and cGAS–STING (Section 6) is an excellent example of how cellular housekeeping processes intersect with DNA damage outcomes.

8. Redox Regulation and NRF2 Interactions with the DNA Damage Response

Radiation's effects are mediated not only by direct DNA breaks but also by a burst of reactive oxygen species (ROS) that cause oxidative damage. Cells mount an antioxidant defense orchestrated by transcription factor **NRF2 (Nuclear Factor E2-related factor 2)**. Intriguingly, recent research has uncovered direct cross-talk between NRF2-driven redox responses and the DNA damage repair/checkpoint machinery:

- **NRF2 and the ATR–CHK1 pathway:** In 2020, Sun *et al.* reported that **NRF2 preserves genomic integrity by facilitating ATR activation and the G2/M checkpoint** [58]. Normally, NRF2's role is to induce antioxidant genes (glutathione synthesis, ROS scavengers) in response to oxidative stress. However, Sun *et al.* found that NRF2 also localizes to chromatin upon radiation and interacts with proteins in the ATR signaling axis. Specifically, NRF2 was shown to cooperate with the ATR activator **TOPBP1** and RPA (which binds ssDNA at stalled replication forks) to enhance ATR–CHK1 signaling [59] [60]. In NRF2-deficient cells, TOPBP1 recruitment to laser-induced DNA damage was impaired and ATR phosphorylation was significantly reduced [61]. Moreover, simultaneous knockdown of NRF2 and TOPBP1 did not further decrease ATR activation compared to either alone, suggesting NRF2 and TOPBP1 function in the same pathway for ATR stimulation [62] [63]. Mechanistically, NRF2 appears to help load TOPBP1 onto chromatin at sites of DNA damage, possibly through a direct or indirect binding (co-immunoprecipitation showed NRF2 and TOPBP1 in a complex on chromatin after irradiation) [64] [61]. As a result, NRF2-proficient cells showed more robust G2 arrest and DNA repair after radiation, whereas NRF2-knockout cells entered mitosis with unrepaired DNA, leading to genomic instability or death. This novel *non-canonical role* of NRF2 means that **NRF2 is not only an antioxidant guardian but also an accessory factor in the DNA damage checkpoint**.

- **NRF2 and radioresistance:** Clinically, tumors with high NRF2 activity (often due to KEAP1 mutations that stabilize NRF2) are notorious for therapy resistance. Part of this is detoxification of ROS, but the findings above suggest NRF2 also raises the threshold for DNA damage by amplifying repair checkpoints. In lung cancer, for instance, NRF2/KEAP1 mutations correlate with poor outcomes to radiotherapy [65] [66]. High NRF2 levels allow cancer cells to better survive oxidative and genotoxic stress – they quickly neutralize ROS and efficiently pause the cell cycle to fix DNA lesions. A recent study by Xiaohui Sun's group extended these findings, showing that **NRF2 can directly bind to TOPBP1 and RPA32 upon radiation, enhancing ATR-mediated phosphorylation of CHK1 and RPA** (thus facilitating replication fork stabilization and DNA repair) [60] [67]. Consistently, **knocking down NRF2 sensitized tumor cells to radiation**, while overexpression made them more resistant by avoiding replication catastrophe.

- **Therapeutic implications:** Targeting the NRF2 pathway could radiosensitize tumors – for example, using brusatol or other NRF2 inhibitors to transiently reduce the antioxidant and ATR-enhancing capacity of cancer cells. Indeed, a study showed that inhibiting NRF2 or its target antioxidant enzymes leads to accumulation of unrepaired DNA damage and increases apoptosis after irradiation [68]. Conversely, radioprotective drugs for normal tissues often activate NRF2 (e.g., dietary antioxidants, pharmacological NRF2 inducers like CDDO-Me or DIM). One striking example is the use of **sulforaphane** (from broccoli sprouts) which activates NRF2; it was shown to mitigate radiation-induced skin injury by reducing oxidative DNA damage and the inflammatory SASP response in keratinocytes [69].

In summary, the **NRF2–DNA repair intersection** exemplifies how oxidative stress and genotoxic stress responses converge. NRF2, through newly discovered interactions with DNA repair proteins, blurs the line between "antioxidant response" and "DNA damage response." It ensures that when radiation generates ROS and DSBs simultaneously, the cell's defenses (antioxidant enzymes and ATR-mediated checkpoints) are mobilized in tandem. Understanding this cross-talk is vital, as it offers new targets (like disrupting NRF2–TOPBP1 binding) to modulate radiosensitivity in cancer versus normal cells [59] [70].

9. Tardigrade "Damage Suppressor" Protein and Extreme Radiotolerance

Tardigrades (water bears) are microscopic animals famous for surviving extreme stresses, including massive doses of radiation. In 2016, a unique tardigrade protein named **"Dsup" (Damage suppressor)** was discovered to protect DNA from ionizing radiation. Continued research in the last few years has deepened our understanding of Dsup and its protective mechanism, with exciting implications for human cells:

- **Dsup binds to chromatin and protects DNA:** Dsup is an intrinsically disordered protein that localizes in the nucleus. Studies using human cells expressing tardigrade Dsup showed a significant reduction in radiation-induced DNA breaks and oxidative damage [71] [72]. In 2017, Chavez *et al.* reported in *eLife* that Dsup physically associates with nucleosomes – it binds DNA and histones – forming a kind of "DNA shield" [73]. **Dsup's binding to chromatin is multivalent and does not block gene transcription, but it sterically protects DNA strands from hydroxyl radicals** produced by radiation [74]. In essence, Dsup acts like a coat over DNA, preventing indirect ROS-mediated strand scission. Structural analyses (published 2022–2023) revealed that Dsup has a DNA-binding region with similarity to nucleosome-binding motifs in other proteins, explaining its affinity for chromatin.

- **Effects of Dsup in vivo and in vitro:** When Dsup is expressed in human cultured cells, those cells exhibit increased radioresistance – fewer DSBs per Gy of radiation and higher survival fractions [71] [72]. Dsup also mitigates other DNA-damaging stresses like H_2O_2 (an oxidizing agent) [71]. A 2023 study by Hashimoto *et al.* in *iScience* extended these findings to multicellular contexts: **Drosophila** and **rice plants** engineered to express Dsup gained improved tolerance to radiation and oxidative stress [75] [76]. In fruit flies, Dsup reduced DNA fragmentation after radiation and improved survival rates [75]. In plants, it conferred protection against genotoxic stress and even drought, hinting at broad stress-shielding properties [76].

- **Mechanistic understanding:** A **2023 Nature Communications** structural study provided refined details of how Dsup works [74]. It appears Dsup can form a network along chromatin, by binding multiple sites, thereby creating a microenvironment that slows diffusion of free radicals to the DNA or directly scavenges them. Dsup does not repair DNA per se; rather it *prevents damage*. In doing so, it reduces the load on classical DNA repair pathways. Tardigrades also have robust DNA repair enzymes, but Dsup gives them a head-start by minimizing initial damage.

- **Prospects for human radioprotection:** The discovery of Dsup has sparked interest in whether its DNA-protective function can be translated to human therapeutics (for radiation protection of normal tissues, or even for protecting cells during space travel). While stable integration of Dsup in humans is far-future, some studies attempt transient expression or peptide derivatives. The concept that *"shielding" proteins* could be used as radioprotectors is new. It also raises evolutionary questions: tardigrades might not have evolved Dsup specifically for radiation, but for desiccation, which also causes DNA damage. Regardless, **Dsup's example teaches us that increasing the physical resilience of chromatin can enhance radiation tolerance in ways complementary to enzymatic repair** [77] [78].

In summary, tardigrade Dsup exemplifies how studying extremotolerant organisms can yield novel radioprotective strategies. It introduced a *previously unknown protein–DNA interaction* that directly safeguards DNA integrity [74]. Dsup's success in human cells [71] is a proof-of-concept that future bioengineering or drug approaches might mimic this mechanism (e.g., small molecules that bind DNA and protect it from radicals). Thus, Dsup has become a symbol of engineered radiotolerance – an idea once in science fiction, now inching toward reality.

10. Mechanisms from Radiodurans: DdrC and DNA Break Stabilization

Another extremophile, the bacterium *Deinococcus radiodurans*, can survive thousands of Gray of radiation, reassembling its shattered genome with astonishing efficiency. Research in the past few years has identified unique proteins in *D. radiodurans* that contribute to its extreme radioresistance. One such factor is **DdrC** (DNA damage response protein C), which illustrates a novel mechanism of dealing with extensive DNA breakage:

- **DdrC senses and stabilizes broken DNA:** A 2023 study by de la Tour and colleagues (published in *Nucleic Acids Research*) found that **DdrC is a DNA damage-induced nucleoid-associated protein that binds to DNA ends and holds them in proximity** [79] [80]. In vivo, DdrC is rapidly recruited throughout the nucleoid of irradiated *Deinococcus* cells, coating the DNA fragments. Biochemical assays showed DdrC can bind both single- and double-stranded DNA. Notably, it appears to promote **chromosomal compaction after damage**, essentially clamping broken pieces together to prevent their dispersion [81]. This "molecular bundling" ensures that fragments originating from the same chromosome remain in vicinity for subsequent annealing and ligation.

- **Lesion-recognition mechanism:** DdrC was shown to have a preference for DNA with irregular geometries (like frayed or damaged ends), suggesting it senses the *presence of a DSB* by structure [82]. Unlike typical eukaryotic repair proteins that accumulate in foci, DdrC spreads across the genome, which may reflect *Deinococcus*' strategy of genome-wide protection. It is akin to forming a temporary protein casing around broken chromosomes. This is reminiscent of the "tethering complex" observed in mammalian cells that holds broken sister chromatids together through mitosis [83], but in bacteria there are no sister chromatids post-irradiation – DdrC likely substitutes by tethering fragments.

- **Facilitating accurate repair:** By compacting the nucleoid, DdrC limits random diffusion of DNA fragments which could lead to misrejoining (translocations). Instead, fragments are corralled so that the efficient *D. radiodurans* homologous recombination machinery (RecA and companions) can find overlaps and stitch the genome back. Indeed, *DdrC knockout bacteria are significantly more radiation-sensitive*, accumulating more chromosomal aberrations. Another protein, **PprA**, previously identified, also binds DNA ends and promotes ligase activity in *Deinococcus*. The network of Ddr proteins (damage response proteins A, B, C, D) upregulated after irradiation includes nucleases, proteases, and now this DNA-end binding protein, all coordinated by a radiation-responsive transcriptional regulator (PprI) [84].

- **Inspiration for human biology:** While bacteria have different DNA organization, the concept of *stabilizing broken DNA collectively* is inspiring new research in eukaryotes. For instance, recent evidence suggests that in human cells with clustered damage (like from high-LET radiation), certain proteins may "glue" multiple breaks together to facilitate repair and prevent chaotic rejoining [85]. Additionally, understanding DdrC's structure may inform engineered proteins that can protect mammalian genomes similarly under massive damage scenarios.

In summary, the mechanisms of *D. radiodurans* highlight that when facing extreme radiation, cells benefit from proteins that *organize and triage DNA damage on a whole-genome scale*. **DdrC's ability to sense and stabilize numerous DNA breaks through nucleoid compaction is a novel mechanism of radioprotection** [81]. It ensures fidelity in the repair of complex damage, something mammals struggle with (leading to chromosomal translocations or loss). By studying DdrC and its allies, scientists hope to uncover strategies to manage heavily fragmented DNA – knowledge that could be relevant to improving outcomes of high-dose radiation exposure or advanced radiotherapies.

11. Novel Radioresistance Factors Revealed by CRISPR Screens

The advent of CRISPR/Cas9 genome-wide knockout (or knockdown) screens has empowered unbiased discovery of genes that modulate cellular radiation sensitivity. Over the past five years, several CRISPR screen studies have identified **previously unappreciated proteins and pathways** that confer radioresistance or radiosensitivity. These findings expand the landscape of radiation biology beyond the "usual suspects" in DNA repair:

- **CARHSP1 – an RNA-binding protein in radioresistance:** A 2021 CRISPR/Cas9 knockout screen in glioblastoma cells pinpointed **CARHSP1 (Calcium-Regulated Heat Stable Protein 1)** as a driver of radioresistance [86]. CARHSP1 is an RNA-binding protein involved in mRNA stability. Knockout of CARHSP1 made glioblastoma cells more vulnerable to radiation, and conversely, overexpression increased survival after radiation. Mechanistically, CARHSP1 appears to stabilize transcripts of stress-response genes during radiation, including cytokines and perhaps DNA repair factors. Its identification is intriguing because it connects post-transcriptional gene regulation with radiation survival – an area not traditionally explored. Targeting CARHSP1 (or its mRNA cargo) could weaken tumor resilience to radiotherapy.

- **Actin-capping proteins and cytoskeletal involvement:** A remarkable discovery from a **2024 bioRxiv preprint by Verma *et al.*** was that **actin cytoskeleton regulators impact radiation response** [87]. Their genome-scale CRISPR screens in cancer cells, including in the context of combined radiation and immunotherapy, found that loss of certain *actin capping proteins* (e.g., CAPZA1/2, CAPZB) significantly sensitized cells to radiation [88]. Actin capping proteins regulate actin filament growth; how do they tie to DNA damage? One hypothesis is that changes in nuclear actin or overall cytoskeletal dynamics affect DNA repair – for instance, through mechanotransduction or by altering nuclear shape and thereby chromatin accessibility. Another possibility is indirect: actin regulators can modulate NF-κB or other survival pathways. This unexpected link opens a new frontier: *the role of cell architecture in genome stability*. It suggests that radiation resistance isn't solely about DNA repair proteins but also about cellular structure and possibly the ability of cells to remodel and recover physically from damage.

- **Other new candidates:** Various screens have yielded hits that point to novel biology: for example, **IRAK1**, a kinase in the IL-1 inflammatory pathway, when depleted made cancer cells more radioresistant [89]. This suggests that pro-inflammatory signaling within tumors might exacerbate radiation killing (or its inhibition helps cells survive). Another CRISPR screen in colorectal cancer identified a microRNA, miR-5197, as a modulator of radiosensitivity [90], hinting at non-coding RNA roles in radiation response. In nasopharyngeal carcinoma, nine new genes (beyond classic repair genes) were found to alter radiosensitivity, including those involved in chromatin remodeling and transcription [91].

- **Synthetic lethal partners with radiation:** CRISPR screens have also been used to find "radiosensitizer targets" – genes that, when knocked out, make radiation far more effective at killing cells. Many hits are in pathways like ATR, ATM, DNA-PK, HR (as expected). But novel ones included metabolic enzymes and cell cycle regulators not previously linked to DDR. This underscores the multi-system nature of radiation response: metabolism, cell cycle, proteostasis, and more all contribute to how a cell negotiates radiation damage.

In summary, **unbiased CRISPR screens have broadened our understanding of radioresistance, revealing new protein–protein interaction networks and pathways that were not traditionally associated with DNA damage repair**. Discoveries like CARHSP1 connect mRNA stability to radiation survival, and actin capping proteins connect cytoskeletal dynamics to DNA repair capacity [88]. These findings not only provide new mechanistic hypotheses (e.g., does the actin network influence nuclear repair factories?) but also identify potential targets to improve radiotherapy. As these screen hits are validated and mechanistically characterized, we expect a more integrated picture of radiation response that spans genome, transcriptome, proteome, and even the cytoskeletal architecture of the cell.

12. Radiation-Induced Bystander and Abscopal Effects via Extracellular Vesicles

Not all effects of radiation are cell-autonomous. The **radiation-induced bystander effect (RIBE)** refers to responses in cells that were not directly hit by radiation but are influenced by signals from irradiated neighbors [92]. Similarly, the **abscopal effect** in cancer therapy is when localized radiation leads to regression of tumors outside the radiation field, presumably through systemic immune signaling. Recent research has zeroed in on the role of **extracellular vesicles (EVs)**, such as exosomes, in mediating these intercellular radiation effects:

- **Exosome-mediated bystander signaling:** Irradiated cells release EVs loaded with a variety of molecular cargo: DNA fragments, RNAs (mRNAs, microRNAs), proteins (including damage-related proteins), and lipids. These exosomes can be taken up by neighboring or distant cells and alter their behavior. For example, it has been shown that exosomes from irradiated cells can cause DNA damage in recipient cells, evidenced by increased γ-H2AX foci in the bystanders [93]. One study demonstrated that *p53 status* of irradiated cells influences their exosomal cargo (via TSAP6, a p53-inducible protein that promotes exosome release) [94] [95]. These radiation-induced exosomes were enriched in certain microRNAs and DNA fragments that, when delivered to bystander cells, triggered DNA damage responses and even apoptosis.

- **Molecular cargo in EVs:** A consistent finding is that irradiated cells' exosomes contain *damage-associated molecular patterns (DAMPs)* and miRNAs that modulate DNA repair gene expression in target cells. For instance, exosomes from X-irradiated human cells carried ATM phosphorylated substrates and bits of mitochondrial DNA, which in new cells activated ATM and NF-κB pathways, leading to oxidative stress and DNA breaks in those cells [96] [97]. MicroRNAs like miR-21, miR-1246 have been identified in radiation-induced exosomes and can suppress specific DNA repair proteins in recipient cells, thereby propagating genomic instability.

- **Epigenetic changes through bystander signals:** There is evidence that bystander cells undergo epigenetic alterations (e.g., DNA methylation changes, chromatin remodeling) due to factors from irradiated cells [98]. Exosomes can deliver enzymes or non-coding RNAs that target epigenetic regulators, potentially explaining the heritable genomic instability observed in progeny of bystander cells.

- **Abscopal effect and immunity:** Exosomes also play a role in systemic immune responses to localized radiation. Tumor-derived exosomes post-irradiation carry tumor antigens and immunostimulatory molecules that can activate dendritic cells and T-cells in lymphoid organs. On the flip side, they may carry immunosuppressive signals as well. For a robust abscopal effect, a balance is tipped towards immunostimulatory content (e.g., calreticulin, HMGB1, type I IFN – which themselves can be secreted free or within EVs). A Nature study in 2020 indicated that **blocking autophagy (see Section 7) led to more mitochondrial DNA in exosomes, which then activated STING-pathway in distant tumors, contributing to abscopal tumor regression** [48]. This underscores how multiple mechanisms tie together: radiation → DNA damage + autophagy changes → exosome cargo changes (more mtDNA) → immune activation in bystander tissues.

- **Clinical implications:** Understanding EVs in RIBE suggests potential mitigators of unwanted bystander damage (e.g., could we filter or modulate circulating exosomes in patients receiving total body irradiation or PRRT?). Conversely, to harness abscopal effects, one might boost the production or the immune-stimulating payload of exosomes from the tumor (some researchers are exploring injecting exosomes from irradiated tumors as a sort of vaccine).

In summary, **intercellular communication via secreted vesicles is a key mechanism by which radiation effects extend beyond the initially hit cells** [96]. The last few years have illuminated the content and consequences of these radiation-induced exosomes: they can carry the "footprint" of radiation damage (DNA DSB signals, DAMPs) and deposit it into otherwise unexposed cells, causing those cells to mount a damage response or even suffer damage themselves [93]. This adds a layer of complexity in radiation biology – the unit of response is not just the individual cell, but a community of cells interconnected by molecular messengers. Managing these bystander signals could improve normal tissue protection and enhance anti-tumor immunity, depending on the context.

13. Genetic Polymorphisms and Radiosensitivity: Radiogenomic Insights (e.g. STING1 Variant)

Why do patients or individuals vary in their sensitivity to radiation? Traditional factors include age, medical co-morbidities, etc., but genetics plays a large role. The field of **radiogenomics** seeks to identify genetic variants associated with adverse effects of radiation therapy or susceptibility to radiation-induced cancers. In the past five years, several genome-wide association studies (GWAS) and meta-analyses have shed light on genetic loci linked to radiation tolerance or injury, highlighting new mechanisms:

- **STING1 locus and mucositis:** A 2022 GWAS in European head-and-neck cancer patients uncovered a significant association between variations on chromosome 5 (in the *STING1* gene) and the risk of developing severe oral mucositis after radiotherapy [99] [50]. The lead SNP (rs1131769) had an odds ratio ~1.95 for mucositis per risk allele [100] [99]. *STING1* encodes the STING protein discussed in Section 6. This finding implies that differences in the innate immune DNA-sensing pathway can affect normal tissue radiosensitivity. It is plausible that certain STING1 haplotypes result in either hyperactive responses (excessive inflammation and tissue damage when radiation triggers STING) or hypoactive responses (less ability to deal with micronucleus DNA, leading to more cell death). The study noted the associated SNP might influence alternative splicing of STING1 [50], though the exact functional consequence requires further investigation. Nonetheless, this is a prime example of a genetic marker pointing to a mechanism: patients with a particular STING variant may experience more inflammation and mucosal injury from the same radiation dose, highlighting the cGAS-STING pathway's role in normal tissue toxicity.

- **Other radiotoxicity loci:** The Radiogenomics Consortium published a meta-GWAS (2019–2020) analyzing tens of thousands of patients across prostate, breast, and other cancer radiotherapy cohorts [101] [102]. They identified loci such as *TANC1* (implicated earlier in fibrosis after prostate RT) and new candidates involved in DNA repair and fibroblast growth factor signaling [103] [104]. For instance, a SNP near **XRCC1** (a base excision repair gene) has been associated with increased risk of late radiation fibrosis in breast cancer survivors, hinting that subtle impairments in DNA single-strand break repair can manifest as long-term tissue damage. Another example is polymorphisms in genes like **SOD2** (antioxidant enzyme) affecting risk of radiation pneumonitis – linking back to the idea that ROS handling capacity (Section 8) matters for tissue tolerance.

- **Heritable DNA repair disorders vs. polymorphisms:** We have long known that rare syndromes (ATM heterozygosity, BRCA mutations, Li-Fraumeni TP53 mutations, etc.) confer hyper-sensitivity to radiation or secondary cancers. The new GWAS findings extend this to common variants with milder effects. They collectively explain a portion of the variance in patient outcomes (e.g., ~29% heritability for acute skin toxicity in one meta-analysis) [105]. These variants often lie in non-coding regions, suggesting regulation of gene expression is key. For example, regulatory variants in **TGFB1** have been linked to risk of fibrosis after RT – consistent with TGF-β being a central driver of tissue fibrogenesis post-radiation.

- **Towards predictive assays:** The ultimate goal is to use a panel of genetic markers to predict who is at higher risk for complications, allowing personalized radiation dosing or added protectants. Already, some studies propose polygenic risk scores for, say, breast fibrosis after RT. Mechanistically, these scores are enriched for genes in **DNA repair, immune response, and extracellular matrix remodeling** (the latter reflecting how tissue healing/scarring processes differ between individuals).

In summary, **radiogenomic studies in the last 5 years have reinforced that individual genetic makeup – often in genes governing DNA repair and immune response – significantly influences radiation sensitivity of normal tissues** [106] [107]. The example of the STING1 polymorphism connecting an innate immune pathway to mucosal injury is especially illuminating [50]. These findings are not only clinically useful but also deepen our mechanistic understanding: they highlight pathways (e.g., innate immune signaling, TGF-β, oxidative damage processing) as crucial in the body's response to radiation injury. Thus, human genetic data complement cellular and animal studies, converging on the same key biological themes of DNA repair capacity and inflammatory control as determinants of radiation tolerance.

14. Influence of the Gut Microbiome on Radiation Tolerance

The **gut microbiome** has emerged as an important factor modulating the response to radiation, especially for the gastrointestinal (GI) tract and systemic immune effects. Research in the past few years has revealed a bidirectional relationship: radiation can cause dysbiosis (altering the microbial community), and conversely the composition of gut microbes can influence how well an organism withstands radiation:

- **Microbiota and GI syndrome:** The gut is one of the most sensitive organs to radiation (hence the GI syndrome at high doses). Recent mouse studies show that radiation causes a loss of beneficial commensals and bloom of opportunistic pathogens in the intestines [108]. This dysbiosis correlates with increased intestinal inflammation, impaired epithelial regeneration, and higher mortality. For instance, *Lactobacillus* species, which produce short-chain fatty acids and support gut barrier integrity, drop after irradiation; their loss is associated with weaker intestinal stem cell recovery and more bacterial translocation leading to sepsis. **Gut microbiota dysbiosis aggravates radiation enteritis by weakening the intestinal epithelial barrier and promoting pro-inflammatory factor expression** [109]**.** Essentially, an unhealthy post-radiation microbiome can turn sublethal intestinal damage into lethal injury through additional inflammatory insult and reduced healing.

- **Microbiome modulation improving outcomes:** Excitingly, **probiotic supplementation or fecal microbiota transplant (FMT)** has shown promise in mitigating radiation injury. For example, feeding mice with *Lactobacillus* or *Bifidobacterium* strains before and after irradiation preserved crypt architecture and reduced chronic inflammation, improving survival. One study demonstrated that a high-fiber diet (which nourishes commensal flora and increases production of butyrate, an anti-inflammatory metabolite) *"boosted radiotherapy efficacy and reduced gut toxicity"* in mouse tumor models [110]. Butyrate can enhance epithelial repair and also modulate the immune response. Another interesting approach was using *Lactobacillus* to metabolize luminal ROS and reactive metabolites generated by radiation, thereby protecting the mucosa.

- **Microbial metabolites and systemic effects:** The microbiome doesn't just act locally in the gut. Metabolites produced by gut microbes can enter circulation and influence bone marrow recovery (e.g., promoting hematopoiesis after radiation) and modulate distant organ inflammation. For instance, **indoles** produced by commensals activate the aryl hydrocarbon receptor (AhR) in intestinal cells, leading to increased resistance to radiation by fortifying tight junctions and producing antimicrobial peptides. Indole-3-carboxaldehyde from *Lactobacillus* was shown to lessen radiation mucositis in mice by this mechanism.

- **Cancer therapy context:** The gut microbiome also affects how patients respond to cancer radiotherapy in terms of tumor control and side effects. Certain microbial signatures correlate with better tumor immune responses (likely by priming immune cells). Clinical trials are exploring if modulating the microbiome (using diet, probiotics, or FMT) can widen the therapeutic window – enhancing tumor kill while protecting normal tissues. One challenge is that broad-spectrum antibiotics given to manage neutropenic infections during chemoradiation can devastate the microbiome and potentially worsen toxicity or reduce abscopal effects.

In essence, **the gut microbiome acts as an "accessory organ" influencing radiation outcomes**. A healthy, fiber-supported microbiota produces factors that reinforce the gut barrier and mitigate oxidative and inflammatory damage from radiation [111] [109]. Conversely, a depleted or imbalanced microbiome exacerbates toxicity. This knowledge pushes radiation biology beyond the host's cells to the symbiotic bacteria within us. It suggests interventions (prebiotics, probiotics, dietary regimens) that could be readily translated to reduce radiation injury in scenarios from radiotherapy to nuclear accidents.

15. Radiation-Induced Cellular Senescence and the SASP

Exposure to ionizing radiation can trigger cells to enter a state of permanent growth arrest known as **cellular senescence**. Senescent cells remain metabolically active and, importantly, secrete a plethora of pro-inflammatory and tissue-remodeling factors termed the **senescence-associated secretory phenotype (SASP)**. Recent research has shed light on the mechanisms linking DNA damage to senescence and how senescent cells influence their microenvironment after radiation:

- **Persistent DNA damage and senescence onset:** If DNA DSBs are misrepaired or remain unrepairable (e.g., telomere damage), they can drive cells into senescence rather than apoptosis. Radiation often causes such lesions, and one hallmark of senescence is the presence of DNA damage foci (53BP1/γH2AX) that persist for days to weeks (sometimes called DNA-SCARS). A key finding was that fragments of DNA can leak from the nucleus of senescent cells into the cytoplasm as **cytoplasmic chromatin fragments (CCFs)** due to nucleus "fragility" [112]. These CCFs activate the cGAS–STING pathway within the senescent cell, leading to a chronic low-level interferon response.

- **cGAS–STING drives SASP:** Ablasser and colleagues discovered that the **cGAS–STING pathway is essential for the full SASP**. In cells induced into senescence by radiation or other stress, cGAS relocation to cytosolic chromatin fragments triggers STING and NF-κB, resulting in secretion of SASP factors like IL-6, IL-8, CXCL10, and HMGB1 [113] [114]. Indeed, **cGAS knockout or STING knockout cells show blunted SASP** – they become senescent (growth-arrested) but do not secrete the high levels of inflammatory cytokines [113] [115]. This was a breakthrough in understanding SASP regulation. It connects the presence of unrepaired DNA damage (and resultant CCFs) to a potent paracrine signaling program. Therefore, in irradiated tissues, senescent cells act as hubs of inflammation partly via ongoing cGAS-STING signaling. This inflammation can recruit immune cells for beneficial clearance of senescent cells, but if senescent cells accumulate (as they often do with age or high radiation doses), the SASP can cause chronic tissue dysfunction (fibrosis, secondary cancer promotion, etc.).

- **Consequences of radiation-induced SASP:** The SASP factors have varied effects. They can reinforce growth arrest in an autocrine manner (stabilizing p53/p16 pathways), and they can impact neighboring cells – for example, IL-6 and IL-8 can induce DNA damage and genomic instability in nearby otherwise healthy cells (a bystander-like effect). TGF-β and IL-1 in SASP drive fibroblast activation and collagen deposition, linking senescence to radiation fibrosis. There's also evidence that SASP factors from senescent endothelial cells contribute to radiation-induced vascular damage and cardiomyopathy.

- **Targeting senescence for radioprotection:** A novel avenue has opened using **senolytics**, drugs that selectively kill senescent cells, to mitigate late radiation effects. For instance, in mouse models of radiation proctitis or pulmonary fibrosis, administering senolytic agents (like a Dasatinib+Quercetin cocktail) after irradiation reduced inflammation and improved tissue function by clearing senescent cells that were fueling SASP-driven damage. Furthermore, compounds that suppress SASP (sometimes called senomorphics) such as **JAK inhibitors** have been shown to alleviate radiation-induced tissue dysfunction without killing the senescent cells.

In summary, **cellular senescence is a fundamental response to radiation damage in normal tissues, and the SASP links DNA damage to long-term tissue environment changes**. The discovery that the cGAS–STING pathway is pivotal for SASP induction provides a molecular mechanism: persistent DNA lesions → chromatin fragments → cGAS/STING activation → NF-κB-driven secretion of inflammatory factors [113] [114]. This mechanistic insight not only enhances our understanding of radiation late effects (like why fibrosis and secondary cancers can occur years after exposure due to a pro-inflammatory milieu) but also suggests interventions (targeting cGAS-STING or SASP factors or senescent cells) to improve outcomes for cancer survivors and radiation accident victims.

16. Radiation-Driven Chromothripsis and Genome Shattering

A relatively recent concept in cancer genomics is **chromothripsis**, a phenomenon where a chromosome (or part of it) shatters into dozens to hundreds of pieces and is haphazardly rejoined, leading to massive rearrangements. While chromothripsis was first observed in tumors (and linked to single catastrophic events, possibly during micronucleus formation or telomere crisis), evidence now implicates *ionizing radiation* as one potential trigger for chromothripsis-like events in cells:

- **Experimental induction of chromothripsis by radiation:** In 2019, a study by Hääg et al. (PLoS ONE) showed that exposing cells (particularly those deficient in TP53) to high-dose gamma radiation can produce karyotypes reminiscent of chromothripsis [116]. They used genome sequencing to demonstrate that certain cells surviving irradiation had clusters of tens of rearrangements confined to one or a few chromosomes – a signature of chromothripsis. Similarly, proton radiation (high linear energy transfer) was reported to cause **chromothripsis-like chromosomal rearrangements** in surviving cells [117]. These findings align with the idea that a single radiation exposure can generate multiple DSBs on a chromosome, and if the cell progresses without proper repair (especially when p53 is mutated and checkpoint is defective), those fragments can reassemble incorrectly in the next cell cycle.

- **Mechanistic basis – micronuclei and misrepair:** How exactly might radiation induce chromothripsis? One hypothesis ties back to micronuclei (Section 6). If a chromosome is damaged and lags in mitosis, it may form a micronucleus. Micronuclear DNA often undergoes erroneous replication and extensive DNA damage, essentially pulverizing that chromosome. When the micronucleus eventually reintegrates into a daughter nucleus, the fragments can be stitched together randomly by NHEJ, yielding a chromothriptic pattern of segmental joins and deletions. Radiation certainly can create the initial chromosome breaks that seed this process. A cell with, say, >10 DSBs in one chromosome has a chance to mis-aggregate and mis-repair them chaotically if checkpoints fail. **Li et al. (2022)** noted that *localized ionizing irradiation within the nucleus induced chromothripsis-like complex alterations* via local clusters of DNA breaks and error-prone rejoining [118] [119].

- **Clinical observations:** Chromothripsis has been observed in secondary malignancies arising after radiation therapy, raising suspicion that the radiation contributed to those complex genomic rearrangements. For example, some radiotherapy-related sarcomas and thyroid cancers bear chromothripsis signatures. It's hard to definitively prove causation in patients, but these correlations, combined with the lab evidence, make a compelling case that radiation can precipitate chromothripsis in vivo. Patients with **Li-Fraumeni syndrome (TP53 mutations)** are especially prone to chromothripsis after radiation, because their cells cannot mount proper DNA damage checkpoints [120] [121].

- **Impact on cells and evolution:** A cell that undergoes chromothripsis either dies (if too much essential material is lost) or, if it survives, it suddenly has a highly rearranged genome, which can spark cancer if it gains a growth advantage or loses tumor suppressors in the process. In tumor evolution, a chromothripsis event can be a shortcut to multi-hit oncogenesis. From a radiation biology perspective, this is the extreme end of mutagenesis – instead of point mutations or simple breaks, you get "genome shattering." Understanding this helps in risk assessment for radiation exposure (the probability of such events might be low, but the consequences are severe and stochastic).

In summary, **chromothripsis represents a catastrophic form of DNA damage outcome that radiation can induce under certain conditions** (especially high doses and compromised checkpoints) [116] [122]. It underscores the importance of proper DNA damage response: robust p53, cell cycle arrest, and accurate repair normally prevent chromosomal chaos, but radiation can overwhelm these safeguards. This area exemplifies the intersection of radiation biology with cutting-edge genomics, illustrating how radiation can accelerate genomic evolution in cells – sometimes giving rise to new deleterious cell lineages (like therapy-related cancers). Ongoing research is investigating biomarkers of chromothripsis in irradiated populations and ways to perhaps prevent extensive misrepair (e.g., enforcing more stringent checkpoints or using means to sequentially repair breaks rather than simultaneously).

17. Non-Canonical Functions of DNA-PKcs Beyond DNA Repair

DNA-PKcs (the DNA-dependent protein kinase catalytic subunit) is well known as a core component of the NHEJ repair pathway for DSBs. However, recent studies have uncovered several **"moonlighting" roles of DNA-PKcs** in cellular physiology, some of which intersect with radiation responses in unexpected ways:

- **DNA-PKcs in innate immunity:** Apart from cGAS-STING, DNA-PKcs itself has been implicated as a DNA sensor in innate immune cells. In dendritic cells and macrophages, DNA-PKcs can bind cytosolic DNA and initiate inflammatory signaling independent of its kinase role in NHEJ [123]. For instance, one study showed that DNA-PKcs positively regulates NF-κB–mediated transcription of cytokines upon detection of foreign DNA or retroelements. *Li et al.* (2016) found that DNA-PKcs interacts with **IRF-3** (an interferon regulatory factor) and promotes IFN-β production in response to DNA virus infection. Therefore, DNA-PKcs contributes to the antiviral immune response; by extension, during radiation (when self-DNA may appear in the cytosol), DNA-PKcs could amplify inflammatory signals. Interestingly, some viruses antagonize DNA-PKcs to avoid this detection.

- **Metabolic regulation by DNA-PKcs:** A surprising development was the link between DNA-PKcs and metabolism, particularly glucose and lipid metabolism. Studies in mice showed DNA-PKcs is involved in insulin signaling and adipogenesis. Nuclear DNA-PKcs can phosphorylate transcription factors like FOXO and hormone-sensitive lipase, influencing metabolic gene expression. From a radiation perspective, metabolic state can influence DDR (e.g., high insulin might dampen some stress responses, caloric restriction might enhance repair). There's evidence that inhibiting DNA-PKcs in obese mice improved metabolic profiles, but how this ties to radiation sensitivity is still being explored.

- **DNA-PK and T cell activation:** DNA-PKcs is abundant in lymphocytes (where its role in V(D)J recombination is crucial). Beyond that, recent work suggests DNA-PKcs has a role in T-cell receptor signaling. A 2020 bioRxiv preprint reported that **DNA-PKcs associates with the LAT signalosome** in T cells to modulate downstream signaling (like PLCγ1 activation) [124]. This means DNA-PKcs might influence how the immune system responds to antigens, including how it responds to tumor antigens after radiation. Inhibiting DNA-PKcs (as done in combining radio- + DNA-PK inhibitor therapies) might have an immunomodulatory effect: it could make tumor cells more sensitive, but also potentially impair T-cell function. However, some data suggest the opposite – DNA-PK inhibition can sustain antitumor immunity by upregulating STING pathway in tumor cells while T cells might compensate via other kinases [125].

- **DNA-PKcs in transcription and telomere maintenance:** DNA-PKcs interacts with RNA polymerase II and certain transcription factors. It's been found at promoters of DNA damage-responsive genes, hinting at a direct role in regulating gene expression post-irradiation. Also, DNA-PKcs is known to localize to telomeres (especially when telomeres become dysfunctional). It can phosphorylate telomeric proteins like TRF2, influencing telomere length maintenance. In the context of radiation, if telomeres are damaged, DNA-PKcs helps prevent end-to-end fusions (a protective role beyond routine DSB repair).

In summary, **DNA-PKcs is not just a DSB repair enzyme; it's a multifunctional protein kinase impacting immune signaling, metabolism, and gene regulation** [123] [126]. This broadens how we think about targeting DNA-PKcs in radiation therapy – inhibitors might radiosensitize tumors by blocking NHEJ, but they also might stimulate STING-dependent immunity (beneficial) or affect T-cells and normal tissue homeostasis (which could be detrimental or beneficial, context-dependent). From a mechanistic standpoint, the non-canonical roles of DNA-PKcs exemplify the interconnectedness of DDR proteins with other cellular networks, reinforcing that the consequences of radiation exposure manifest at multiple biological levels.

18. Replication Stress, ATR, and RPA Exhaustion in Irradiated Cells

Radiation exposure can induce **replication stress** – especially when cells in S-phase experience DNA damage that stalls replication forks. Over the last few years, significant advances have been made in understanding how replication stress is managed and how it interfaces with radiation-induced damage:

- **Concept of RPA exhaustion:** When replication forks stall or collapse (e.g., due to single-stranded DNA from a nick or an unrepaired lesion), stretches of single-stranded DNA (ssDNA) are coated by Replication Protein A (RPA). If many forks stall simultaneously (which can happen after radiation causing multiple SSBs and DSBs), the cellular pool of RPA can be titrated out – a situation termed *RPA exhaustion*. Recent studies showed that excessive ssDNA generation can sequester RPA and activate ATR in a panicked, global fashion [127]. ATR signaling will attempt to stabilize forks and halt the cell cycle (ATR–CHK1–WEE1 pathway) [128], but if RPA is fully consumed, new ssDNA regions go unprotected, leading to fork collapse and DSB formation. **In essence, uncontrolled replication stress can convert into secondary DSBs – compounding the initial radiation damage**. This provides a mechanistic understanding of the long-observed synergy between radiation and S-phase: cells irradiated in S-phase often suffer replication-associated DSBs beyond the immediate DNA breaks.

- **ATR's role in radioresistance:** ATR kinase has emerged as a critical protector against replication stress during radiation exposure. It prevents extensive origin firing (via CHK1 and WEE1) and stabilizes stalled forks, giving time for repair [127]. ATR inhibitors are being tested clinically as radiosensitizers, precisely because they remove this safety net, forcing replication forks to collapse in irradiated tumor cells. Conversely, in normal cells, strong ATR activation helps them cope with moderate radiation by pausing S-phase or slowing replication to allow repair.

- **Interplay of replication and repair pathways:** Recent research highlights cooperation between replication stress responders and DSB repair factors. For example, **RAD51** (an HR protein) has a replication fork protection role – it helps to remodel and restart stalled forks. After radiation, RAD51 is loaded not only at DSBs but also at stressed forks to prevent breakage. Also, the **Fanconi Anemia (FA) pathway**, known for repairing DNA interstrand crosslinks, is activated by radiation-induced replication stress to stabilize forks and channel lesions into HR. Loss of FA components makes cells particularly sensitive to the combination of radiation and replication (e.g., during fractionated therapy, FA-mutant cells accumulate chromosomal breaks due to inability to resolve stalled forks).

- **Mitotic catastrophe if replication stress is unmitigated:** Another connection: if replication stress is excessive and ATR/CHK1 fail to maintain the G2/M checkpoint, cells may enter mitosis with under-replicated DNA, leading to broken chromosomes in mitosis and mitotic catastrophe. Polθ's newly found role in mitotic DSB repair (Section 2) is like a last-ditch effort to rescue such situations [9]. Understanding this, some researchers propose scheduling radiotherapy doses to specific cell cycle phases or using CDK inhibitors to manipulate S-phase entry, thereby controlling replication stress during treatment.

In summary, the convergence of **replication stress and radiation damage** research has yielded the concept that a cell can survive radiation better if it effectively manages the DNA replication process afterwards [127]. ATR and RPA are linchpins in this process: they sense the ssDNA, signal to slow cell cycle, and coordinate fork repair [127]. RPA exhaustion is a tipping point – once RPA is depleted, fork collapse and DSBs skyrocket. This explains why inhibitors of key S-phase checkpoint proteins (ATR, CHK1, WEE1) dramatically sensitize cells to radiation – they induce an unsustainable replication stress leading to lethal DNA damage [127]. It's a beautiful example of exploiting a mechanistic weakness: the dependency of damaged cells on replication stress responses. From a fundamental perspective, it underscores that the temporal sequence of damage (DSBs) and ongoing replication can dictate cell fate, reaffirming the old radiobiological tenet that *cycling cells are more radiosensitive*, now with molecular detail.

19. Chromatin Modifications and DNA Repair Pathway Choice

Chromatin state profoundly influences DNA repair pathway selection. In recent years, particular post-translational modifications of histones have been identified as *molecular switches* that channel DSBs into either NHEJ or homologous recombination (HR). Two modifications have garnered special attention: **histone H4 lysine 20 methylation (H4K20me)** and **histone H3 lysine 36 methylation (H3K36me)**:

- **H4K20 methylation and 53BP1:** It was known that 53BP1, a pro-NHEJ factor, binds to dimethylated H4K20 on nucleosomes near DSBs. Recent work has further clarified that the *absence* of H4K20me2 (as occurs in newly replicated chromatin where H4K20 is transiently unmethylated or only monomethylated) is a signal favoring HR, whereas constitutive H4K20me2 (found in heterochromatin and G1 cells) attracts 53BP1 to block resection and favor NHEJ. In 2019, Nakamura *et al.* showed that cells lacking the enzyme SET8 (which monomethylates H4K20) had repair defects because they couldn't properly recruit 53BP1 [129]. Conversely, overabundance of H4K20me2 can overly repress resection. Thus, writers and erasers of H4K20 methylation (SET8, SUV4-20, etc.) are crucial in modulating radiation response by tilting repair toward NHEJ or HR.

- **H3K36 methylation (me2 vs me3) and pathway choice:** A significant advance was the realization that different methylation states of H3K36 guide distinct DSB repair pathways [130]. H3K36me3, typically found in actively transcribed gene bodies (euchromatin), was linked to HR. Mechanistically, H3K36me3 is recognized by a protein called **LEDGF (p75)** which in turn recruits the CtIP/Sae2 nucleases to promote DNA end resection – a prerequisite for HR. On the other hand, **H3K36me2**, more enriched in intergenic or heterochromatic regions, was found to recruit the **NuMA protein** (via its reader ZMYND11) and favor NHEJ. A 2020 study by Carvalho *et al.* showed **H3K36me2 and H3K36me3 have distinct roles: H3K36me2 biased towards 53BP1 loading (NHEJ), whereas H3K36me3 promoted BRCA1 and RAD51 loading (HR)** [130]**.** The cell can dynamically regulate the conversion of me2 to me3 through enzymes like NSD1/2 (for me2) and SETD2 (for me3). In S-phase, when HR is most needed, SETD2 activity ensures high H3K36me3 at DSBs, fostering HR [131]. If SETD2 is lost (as in some cancers), those cells rely more on NHEJ and become sensitive to PARP inhibitors (an interesting synthetic lethal scenario).

- **Other histone marks:** Histone acetylation also matters. For instance, H4 acetylation (H4K16ac) relaxes chromatin and was found to enhance 53BP1 recruitment by exposing H4K20me2. An enzyme SIRT1 (a deacetylase) removes H4K16ac near breaks to enable 53BP1 binding – if SIRT1 is inhibited, HR is favored. Similarly, macroH2A histone variant incorporation can repress HR in some contexts. And the histone variant H2AX, when phosphorylated to γH2AX, creates a landing pad for MDC1 and the subsequent cascade of ubiquitin signaling that is essential for both HR and NHEJ factor accrual.

- **Translational insight:** Modulating chromatin modifiers could influence radiosensitivity. For example, HDAC inhibitors (which increase histone acetylation and generally loosen chromatin) tend to interfere with NHEJ and enhance DNA damage (hence their use as radiosensitizers). In contrast, inhibitors of demethylases that remove H3K36me3 might push repair toward error-prone NHEJ, which could be exploited in tumor cells to induce more misrepair. One can envision personalized radiotherapy where, if a tumor has a particular chromatin dysregulation (e.g., SETD2 mutation, meaning low H3K36me3 and HR deficiency [130]), one might choose particular DDR inhibitors to exacerbate this, or conversely, protect normal tissues by temporarily adjusting chromatin state (perhaps via dietary factors or epigenetic drugs).

In conclusion, **chromatin marks serve as a code that instructs the cell which DNA repair pathway to employ, thereby influencing the cell's fate after radiation** [130]. The interplay of H4K20me and H3K36me modifications with 53BP1 and BRCA1/HR factors is a prime example: it's a molecular decision point that can tilt the balance between quick-and-dirty repair (NHEJ) and slow-but-accurate repair (HR). By deciphering this code, we not only gain fundamental insight into genome stability maintenance but also open avenues for tweaking these signals to favor outcomes we want (like better repair in normal cells, or more lethal damage in cancer cells).

20. Immunogenic Cell Death and DAMP Signaling After Radiation

Radiation can induce a form of cell death that is particularly visible to the immune system, known as **immunogenic cell death (ICD)**. In ICD, dying cells release or expose certain molecules that serve as *danger signals*, or DAMPs (damage-associated molecular patterns). The concept of radiation-induced ICD has gathered strong support, especially in how radiotherapy can synergize with immunotherapies. Key features and mechanisms include:

- **Key DAMPs from irradiated cells:** The classic hallmarks of ICD are: cell-surface exposure of *calreticulin* (CRT), release of *ATP* and *high mobility group box 1* (HMGB1), secretion of *type I interferons*, and release of *Annexin A1*, among others [132]. Radiation can trigger all these: for instance, pre-apoptotic cells expose calreticulin on their membrane (acting as an "eat me" signal for dendritic cells), and post-apoptotic cells passively release HMGB1 from the nucleus (which then binds Toll-like receptor 4 on immune cells to promote antigen presentation) [132]. The timing of these signals is crucial – calreticulin exposure often precedes phosphatidylserine exposure in apoptosis; radiation seems particularly good at inducing that early CRT exposure, especially in combination with certain chemotherapies or ATR inhibitors [133] [134].

- **Dose and context dependence:** Studies indicate that higher dose per fraction (e.g., stereotactic radiation) is more efficient at inducing ICD than conventional low-dose fractionation [135]. High dose leads to extensive DNA damage which drives robust type I IFN production via cGAS-STING, an important component of ICD (as IFN activates dendritic cells and facilitates cross-priming of T-cells). For example, *carbon ion radiation* at a single high dose was shown to induce classic ICD markers (CRT exposure, ATP release) in a murine tumor model, leading to superior anti-tumor immune responses compared to X-rays [136] [137]. This has spurred interest in the synergy between hypofractionated RT and immune checkpoint blockade in the clinic.

- **Caspases and ICD regulation:** Interestingly, apoptosis executed purely through the intrinsic pathway (e.g., caspase-9 and -3 activation) can sometimes be less immunogenic because it rapidly dismantles the cell without giving time for DAMP signaling. There is evidence that caspase inhibitors or knockdowns can paradoxically increase emission of certain DAMPs. One study noted that *tumor cells suppress radiation-induced immunity by hijacking caspase-9 signaling*, suggesting that modulating the cell death modality (apoptosis vs. necroptosis vs. mitotic catastrophe) changes immunogenicity [138]. Radiation in combination with agents that skew cell death toward necroptosis or a pro-inflammatory form of apoptosis could enhance ICD.

- **Biomarkers and clinical translation:** Researchers are seeking biomarkers of RT-induced ICD in patients. For example, elevated HMGB1 in serum post-RT, or increased IFN-stimulated gene expression in blood, could indicate an ongoing abscopal immune effect [139]. Trials combining radiotherapy with immunotherapy often measure such markers. Early-phase trials have reported that patients whose tumors showed more calreticulin and PD-L1 upregulation after radiation had better responses when given anti-PD-1 therapy, supporting the idea that radiation can convert tumors into in situ vaccines in some cases.

In summary, **radiation can provoke an immunogenic mode of cell death wherein dying tumor cells broadcast “come eat me” and “danger” signals to the immune system** [132]. Key molecular interactions have been elucidated: calreticulin on the tumor cell binds to CD91 on dendritic cells, ATP engages P2X7 receptors on dendritic cells promoting inflammasome activity, and HMGB1 stimulates TLR4 on antigen-presenting cells [132] [134]. These signals result in efficient uptake of tumor antigens by dendritic cells and cross-priming of CD8 T cells, thereby linking the local effect of radiation to a systemic anti-tumor response. This paradigm has reinvigorated interest in using radiotherapy not just for direct tumor cytotoxicity but as a tool for immune modulation. Mechanistically, it underscores that cell death is not a silent process – the way cells die (and what they release in the process) profoundly influences the outcome for the organism, especially in the context of cancer therapy.

Conclusion: The past five years have significantly enriched our understanding of radiation biology, moving the field well beyond the classical DNA damage and repair framework. We now appreciate that radiation triggers a web of interconnected processes: sophisticated new protein complexes (Shieldin, Polθ/TMEJ machinery) ensure DNA ends are handled correctly; dynamic phase-separated condensates organize the repair milieu; RNA molecules and RBPs actively participate in the damage response; innate immune sensors like cGAS-STING link DNA damage to inflammation; autophagy and the microbiome influence how tissues cope with damage; chromatin context and histone modifications dictate repair pathway choice; and on a higher level, intercellular and systemic signals (exosomal communication, immunogenic cell death) determine organismal responses to radiation. These 20 novel ideas and discoveries underscore a recurring theme: **the response to radiation damage is multi-dimensional**, involving genome, epigenome, transcriptome, proteome, organelles, intercellular networks, and even commensal organisms. For experts in radiation research, integrating these insights will be crucial for devising better radioprotectors, radiosensitizers, and therapeutic strategies. The rigor of mechanistic detail now available – much of it gleaned from primary literature and preprints we have cited – provides a strong foundation for translating these ideas into clinical gains, while also inspiring further fundamental questions about how life endures (or succumbs to) the stress of radiation.

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